AMAN AHMED

+1-510-565-9897 | amanahmed212@gmail.com | amanahmed212 | aman1678 | San Francisco Bay Area, CA

U.S. Citizen

EDUCATION

- **University of California, San Diego** *Graduated June 2025*
  B.S. in Physics with Honors La Jolla, CA

WORK EXPERIENCE

- **Full Stack Developer (Freelance)** *February 2025 – June 2025*
  Qanvas (UC San Diego Startup) La Jolla, CA
  - Built a full-stack web platform with a **React** and **JavaScript** frontend and a **Flask** backend, bundled with **Vite**.
  - Designed a **REST API** with 10+ endpoints for data ingestion, user management, and simulation dispatch.
  - Engineered async data pipelines in **Python** that validated, normalized, and stored data in JSON format.
  - Developed a modern **UI/UX** tailored towards client base, including new features unavailable in alternative applications.

PROJECTS

- **Real-Time Gravitational Simulation Engine**
  Python, NumPy, Matplotlib, Pygame
  - Architected a real-time multi-body physics engine in **Python**, with the simulation core, rendering layer, and user controls separated into independent modules using object-oriented design.
  - Implemented **numerical integration** for orbital dynamics and designed **NumPy**-vectorized adaptive approximation algorithms to improve both accuracy and runtime performance.
  - Validated output by comparing simulated trajectories against expected physical behavior in **Matplotlib**, diagnosing and correcting accumulated numerical drift.
  - Built an interactive real-time front end that renders 2D trajectories and lets users adjust the simulation while it runs.

- **Driew – Pakistani Drama Review Platform**
  React, TypeScript, Vite, Python, Flask, Flask-SQLAlchemy, SQLite
  - Building Driew, a full-stack web app where users browse and share reviews and discussions about Pakistani dramas, with a **React** and **TypeScript** frontend bundled with **Vite**.
  - Developed a **Python** and **Flask REST API** exposing endpoints to create, view, edit, and delete posts, returning JSON for the frontend to consume.
  - Modeled post data as relational models with **Flask-SQLAlchemy** and persisted it in a **SQLite** database, so posts remain available across sessions and server restarts.
  - Added post management features that let users update post descriptions after publishing, handled through typed **TypeScript** API calls to the Flask backend.

- **Cowboy Apocalypse**
  Python, Pygame
  - Engineered a real-time 2D game simulation in **Python** and **Pygame**, modeling the player, enemies, projectiles, and platforms as encapsulated classes across separate sprite, utility, and game-loop modules.
  - Built a 60 FPS update–render loop with **delta-time integration**, keeping movement, gravity, and projectile speed frame-rate independent.
  - Implemented **collision detection** between entities, platforms, and projectiles, plus enemy chase behavior and per-entity health and damage state.
  - Managed time-based state (fire-rate cooldowns, magazine capacity, timed reloads) and an event-driven audio and HUD system for live player feedback.

SKILLS

- **Programming:** Python, SQL, TypeScript, HTML, CSS, JavaScript
- **Libraries and Frameworks:** NumPy, Pandas, SciPy, Matplotlib, React, Flask, Node.js
- **Tools & Methods:** Git, Bash, Claude, GenAI Fluency, Vite, Vercel, SQLite, RESTful APIs, SQLAlchemy, CRUD
- **Languages:** Urdu (Native/Bilingual Fluency), Hindi (Native/Bilingual Fluency)

HONORS

- **Tau Sigma National Honors Society** *February 2024*
  - Recognized for academic excellence and leadership among transfer students.